Revision Session 1

Microeconomics

with step-by-step solutions

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PART 1

Multiple-choice questions

Question 1(i)

Varian, Chapter 1

Pareto

"A change that makes at least one person better off without making anybody worse off". This is the definition of:

(a) An equitable reform
(b) A productive variation
(c) Pareto efficiency
(d) A Pareto improvement

Question 1(i) – step by step

Step 1. Key word: the definition describes a *change*, i.e. a move from one allocation to another.

Step 2. A change that makes someone better off and nobody worse off is a **Pareto improvement** $\Rightarrow$ **(d)**.

Step 3. Discard (c): *Pareto efficiency* describes an *allocation*, not a change. An allocation is Pareto efficient when no Pareto improvement is possible from it.

Step 4. Discard (a) and (b): they are not technical terms of the course.

Typical mistake: confusing (c) and (d). The two concepts are linked (efficient = no improvement left), but they are not the same object.

Question 1(ii)

Varian, Chapter 3

Properties of preferences

Suppose that Barney's musical tastes are such that he prefers Jazz music to Rock music, Rock music to Classical music and Classical music to Jazz music. Barney's preferences do not display:

(a) Reflexivity
(b) Transitivity
(c) Completeness
(d) Both (a) and (b)
(e) All of (a), (b) and (c)

Question 1(ii) – step by step

Step 1. Write the preferences: $J \succ R$, $R \succ C$, $C \succ J$.

Step 2. **Transitivity**: if $J \succ R$ and $R \succ C$, then it must be $J \succ C$. But Barney has $C \succ J \Rightarrow$ transitivity is **violated**.

Step 3. **Completeness**: every pair can be compared. All three pairs (J, R), (R, C), (C, J) are ranked $\Rightarrow$ satisfied.

Step 4. **Reflexivity**: each bundle is at least as good as itself. Nothing in the question contradicts it $\Rightarrow$ satisfied.

Step 5. Only transitivity fails $\Rightarrow$ **(b)**.

Why it matters: with a cycle there is no "best" option among the three, so the choice problem has no solution.

Question 1(iii)

Varian, Chapter 2

Budget line

Suppose there are two goods, ***x*** and ***y***, the prices of both goods are positive and a consumer's income is also positive. Suppose that the prices of good ***x*** and good ***y*** both double and income triples. On a graph where the budget line is drawn with the quantity of ***x*** on the horizontal axis and the quantity of ***y*** on the vertical axis,

(a) the budget line becomes steeper and shifts outward.
(b) the budget line becomes flatter and shifts outward.
(c) the new budget line is parallel to the old budget line and lies below it.
(d) the new budget line is parallel to the old budget line and lies above it.
(e) None of the above.

Question 1(iii) – step by step

Step 1. Initial budget line: $p_x x + p_y y = m$.

Step 2. New budget line: $2p_x x + 2p_y y = 3m$. Divide by 2:

$$p_x x + p_y y = 1.5\,m$$

Step 3. Slope: $-p_x/p_y$ before, $-2p_x/2p_y = -p_x/p_y$ after $\Rightarrow$ unchanged, the two lines are **parallel**.

Step 4. Intercepts: $m/p_x \to 1.5\,m/p_x$ and $m/p_y \to 1.5\,m/p_y \Rightarrow$ the line moves **outward**.

Step 5. Parallel and above $\Rightarrow$ **(d)**.

Rule: only *relative* prices and *real* income matter. Multiplying all prices by t and income by s is equivalent to multiplying income by s/t (here $3/2$).

Question 1(iv)

Varian, Chapter 6

Types of goods

"A good for which demand increases as income falls." This is the definition of:

(a) a Normal good
(b) an Ordinary good
(c) an Inferior good
(d) a Giffen good

Question 1(iv) – step by step

Step 1. The definition is about a change in **income**, with prices fixed. So the candidates are the two income-based categories: normal and inferior.

Step 2. Demand rises when income falls: $\Delta x/\Delta m < 0 \Rightarrow$ **inferior good**, **(c)**.

Step 3. Discard (a): normal means $\Delta x/\Delta m > 0$.

Step 4. Discard (b) and (d): ordinary and Giffen are defined by the response to the good's own **price**, not to income (ordinary: demand falls when the price rises; Giffen: demand rises when the price rises).

Typical mistake: answering (d). Giffen goods are inferior (see 1(v)), but the definition in the question does not mention prices.

Question 1(v)

Varian, Chapters 6 and 8

Inferior and Giffen goods

Which of the following statements is correct?

(a) Giffen goods are always inferior goods
(b) Inferior goods are always Giffen goods
(c) Not all inferior goods are also Giffen
(d) (a) and (b) are both correct
(e) (a) and (c) are both correct

Question 1(v) – step by step

Step 1. Slutsky decomposition of a price increase:

$$\Delta x = \underbrace{\Delta x^{s}}_{\text{substitution}} + \underbrace{\Delta x^{n}}_{\text{income}}$$

Step 2. The substitution effect always goes against the price: price up $\Rightarrow \Delta x^{s} \leq 0$.

Step 3. Giffen: price up $\Rightarrow \Delta x > 0$. This needs an income effect that is *positive* and *larger* than the substitution effect. A price increase lowers purchasing power, so a positive income effect means the good is inferior $\Rightarrow$ (a) is **true**.

Step 4. The converse fails: an inferior good whose income effect is smaller than the substitution effect still has $\Delta x < 0$, so it is ordinary, not Giffen $\Rightarrow$ (b) is **false**, (c) is **true**.

Step 5. (a) and (c) are correct $\Rightarrow$ **(e)**.

In one line: Giffen $\subset$ inferior, and the inclusion is strict.

Question 1(vi)

Varian, Chapter 9

Labour supply

If leisure is a normal good, then an increase in income that does not involve a change in the wage rate (e.g. an increase in non-labour income) will tend to result in:

(a) An increase in labour supply
(b) A fall in labour supply
(c) No change in labour supply
(d) Any of (a), (b) or (c) are equally likely

Question 1(vi) – step by step

Step 1. The wage, i.e. the price of leisure, does not change $\Rightarrow$ the slope of the budget line does not change. The line shifts **up**, parallel.

Step 2. So there is only an **income effect**, with no substitution effect.

Step 3. Leisure is normal $\Rightarrow$ more income, more leisure: R rises.

Step 4. Time is fixed: labour $=$ total hours $- R \Rightarrow$ labour supply **falls**, **(b)**.

Contrast with a wage increase: there the substitution effect (leisure becomes more expensive) and the income effect go in opposite directions, and the sign of the change in labour supply is ambiguous.

PART 2

Cobb–Douglas demand

Exercise 2

Varian, Chapter 5

Tracey's choice

Tracey's utility function between consumption of heating fuel (bought in quantity f) and consumption of other goods (quantity g) may be described by

$$u(f,g) = f^{0.5}g^{1.5}$$

So marginal utilities are

$$MU_f = \frac{\partial u}{\partial f} = 0.5\,f^{-0.5}g^{1.5} \qquad \text{and} \qquad MU_g = \frac{\partial u}{\partial g} = 1.5\,f^{0.5}g^{0.5}$$

and her budget is a fixed money income m.

(a) Denote the price of fuel by P_f and the price of other goods by P_g. What is the equation for Tracey's budget constraint?

(b) Using the condition for optimality that relates the marginal rate of substitution and prices, derive an expression for the relationship between (optimally chosen) expenditure on other goods and (optimally chosen) expenditure on fuel.

Exercise 2 (continued)

Varian, Chapter 5

Tracey's choice

(c) Solve for the demand functions for ***f*** and ***g***.

(d) Suppose Tracey is observed at three points in time and the only things that change between points in time are Tracey's income level and prices. At the points in time, Tracey's expenditure on fuel and her total income are observed to be:

	Expenditure on fuel	Total income
Time 1	25	100
Time 2	30	120
Time 3	20	80

Are these expenditure patterns consistent with the utility function proposed in the question?

Exercise 2(a)–(b) – step by step

(a) Step 1. Expenditure on fuel plus expenditure on other goods equals income:

$$P_f f + P_g g = m$$

(b) Step 1. Marginal rate of substitution:

$$|MRS| = \frac{MU_f}{MU_g} = \frac{0.5 f^{-0.5} g^{1.5}}{1.5 f^{0.5} g^{0.5}} = \frac{0.5}{1.5} \cdot \frac{g^{1.5-0.5}}{f^{0.5+0.5}} = \frac{g}{3f}$$

Step 2. At the optimum the indifference curve is tangent to the budget line:

$$\frac{g}{3f} = \frac{P_f}{P_g}$$

Step 3. Cross-multiply:

$$\boxed{P_g g = 3 P_f f}$$

Optimal expenditure on other goods is three times optimal expenditure on fuel.

Signs: Varian writes $MRS = -MU_f/MU_g = -P_f/P_g$. Same condition, both sides negative.

Exercise 2(c) – step by step, method 1

Two equations in two unknowns

Step 1. The system:
$$\begin{cases} P_f\, f + P_g\, g = m & \text{(budget constraint)} \\ P_g\, g = 3\, P_f\, f & \text{(tangency, from (b))} \end{cases}$$

Step 2. Substitute the second equation into the first:
$$P_f\, f + 3\, P_f\, f = m \quad \Longrightarrow \quad 4\, P_f\, f = m$$

Step 3. Solve for f:
$$\boxed{f = \frac{m}{4P_f} = 0.25\,\frac{m}{P_f}}$$

Step 4. Plug back into the tangency condition: $P_g\, g = 3P_f \cdot \dfrac{m}{4P_f} = \dfrac{3m}{4}$, so
$$\boxed{g = \frac{3m}{4P_g} = 0.75\,\frac{m}{P_g}}$$

Step 5. Check: $P_f f + P_g g = m/4 + 3m/4 = m$. ✓

Exercise 2(c) – step by step, method 2

The Cobb–Douglas shortcut

Step 1. With $u(x_1, x_2) = x_1^c x_2^d$, the consumer spends a fixed share of income on each good:

$$p_1 x_1 = \frac{c}{c+d}\, m \qquad p_2 x_2 = \frac{d}{c+d}\, m$$

Step 2. Here $c = 0.5$, $d = 1.5$, $c + d = 2$:

$$\text{share of fuel} = \frac{0.5}{2} = 0.25 \qquad \text{share of other goods} = \frac{1.5}{2} = 0.75$$

Step 3. So $P_f f = 0.25\, m$ and $P_g g = 0.75\, m$, which gives the same demand functions:

$$f = 0.25\, \frac{m}{P_f} \qquad g = 0.75\, \frac{m}{P_g}$$

The exponents do not need to sum to 1. $u^{1/2} = f^{0.25} g^{0.75}$ is a monotonic transformation of u: it represents the same preferences and makes the shares visible directly in the exponents.

Exercise 2(d) – step by step

Step 1. What the model predicts: from (c), $P_f f = 0.25\,m$. The share of income spent on fuel is **25%**, whatever the prices and whatever the income.

Step 2. Compute the observed shares:

$$\frac{25}{100} = 0.25 \qquad \frac{30}{120} = 0.25 \qquad \frac{20}{80} = 0.25$$

Step 3. The share is 25% at all three points in time $\Rightarrow$ **yes**, the data are consistent with $u(f, g) = f^{0.5} g^{1.5}$.

Step 4. Note: we do not need to know the prices. With Cobb–Douglas preferences, expenditure shares do not depend on prices.

"Consistent" is not "proven": any Cobb–Douglas utility with $c/(c + d) = 0.25$ (e.g. $f^1 g^3$) fits the same data.

PART 3

Labour supply with a benefit and a tax

Exercise 3

Varian, Chapter 9

Labour–leisure choice

Suppose that an individual has 20 hours a day (!) available to allocate between work (***L***) or leisure (***R***). Suppose that he receives a wage rate of 10 dollars an hour, and that the other good that provides him with utility is consumption spending (***C***), measured in dollars.

(a) Draw the budget line for this consumer's problem, putting ***R*** on the horizontal axis and being careful to label axes and the intercepts of the curve.

Exercise 3 (continued)

Varian, Chapter 9

A benefit and a tax

Now suppose that the government introduces a benefit payment that boosts income by 20 dollars per individual per day. To pay for this benefit, the government introduces a tax which takes 20 cents of each dollar of earned income and which starts being paid once labour income (i.e. income excluding the 20 dollars of benefit) exceeds 20 dollars per day.

(b) At what number of hours of work will the tax start to apply for the individual considered in this question?

(c) Including the effects of the benefit payment and of the tax, what will be the individual's income if he works: (i) 0 hours; (ii) 5 hours; (iii) 10 hours; (iv) 20 hours.

(d) On your graph for part (a), draw a new budget line that represents the case with the benefit and the tax both affecting income. Be careful to label axes, the intercepts of the curve, and the point at which the tax starts to apply.

Exercise 3(a) – step by step

Step 1. Time constraint: $L + R = 20$, so $L = 20 - R$.

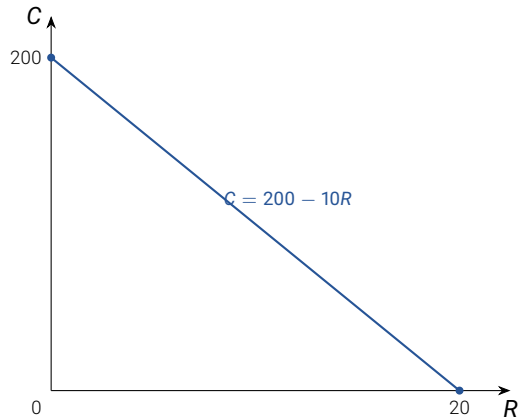
Step 2. Consumption equals labour income:

$$C = 10\,L = 10\,(20 - R) = 200 - 10R$$

Step 3. Intercepts:

- $R = 0$ (works 20 hours): $C = 200$
- $R = 20$ (does not work): $C = 0$

Step 4. Slope: $-10 = -w$. One more hour of leisure costs 10 dollars of consumption.



Exercise 3(b)–(c) – step by step

(b) Step 1. The tax starts when labour income exceeds 20 dollars: $10L > 20 \iff L > 2$. The tax starts to apply after **2 hours** of work ($R = 18$).

(c) Step 1. Write total income Y in two pieces:

$$Y = \begin{cases} 20 + 10L & L \le 2 \\ 20 + 10L - 0.2\,(10L - 20) = 24 + 8L & L > 2 \end{cases}$$

Step 2. Only the labour income *above* 20 dollars is taxed.

Hours	Labour income	Tax	Benefit	Total income
0	0	0	20	**20**
5	50	$0.2 \times 30 = 6$	20	**64**
10	100	$0.2 \times 80 = 16$	20	**104**
20	200	$0.2 \times 180 = 36$	20	**184**

Check with the formula: $24 + 8 \times 5 = 64$, $24 + 8 \times 10 = 104$, $24 + 8 \times 20 = 184$.

Exercise 3(d) – step by step

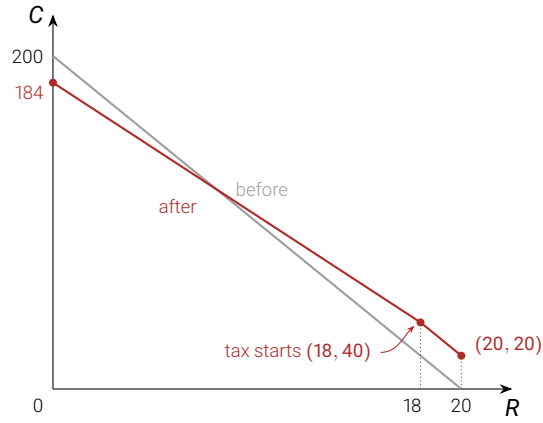
Step 1. Rewrite the income formula in terms of $R = 20 - L$:

- $18 \le R \le 20$: $C = 220 - 10R$, slope -10
- $0 \le R \le 18$: $C = 184 - 8R$, slope -8

Step 2. Key points:

- no work ($R = 20$): $C = 20$, the benefit only
- kink at $R = 18$ ($L = 2$): $C = 40$
- $R = 0$ ($L = 20$): $C = 184$

Step 3. The new line **no longer intercepts the R axis**: it ends at $(20, 20)$. It intercepts the C axis at 184, and above the kink it is flatter, because the net wage is $0.8 \times 10 = 8$.



Exercise 3 – reading the graph

Step 1. Where do the two budget lines cross? Set the two consumption levels equal on the segment above the kink:

$$184 - 8R = 200 - 10R \quad \Longrightarrow \quad 2R = 16 \quad \Longrightarrow \quad R = 8 \;\; (L = 12), \quad C = 120$$

Step 2. Below 12 hours of work ($R > 8$) the new line lies **above** the old one: the benefit (20) is worth more than the tax paid. Check at $L = 5$: 64 against 50.

Step 3. Above 12 hours of work ($R < 8$) the new line lies **below**: the tax paid exceeds 20. Check at $L = 20$: 184 against 200.

Step 4. At exactly 12 hours the tax paid is $0.2 \times (120 - 20) = 20$, equal to the benefit: the individual is unaffected.

Link with Question 1(vi): for someone who works few hours, the benefit is mostly non-labour income (income effect only); above the kink the net wage falls from 10 to 8, which also adds a substitution effect.